Introduction to Python Data Structures: Lists, Tuples, and Dictionaries

Python Basics Notes

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Lists, tuples, and dictionaries are Python's three most fundamental built-in data structures used to group multiple items together under a single variable name.

1 List

An ordered, flexible collection defined using square brackets `[]`.

- **Mutable:** You can freely add, remove, or modify items after creation.
- **Indexed:** Items are retrieved using their numerical position, starting from `0`.
- **Best used for:** Collections of similar items that change over time (e.g., a shopping list, to-do tasks).

```
shopping_list = ["apples", "milk", "bread"]
shopping_list.append("eggs")       # Adds a new item
shopping_list[0] = "green apples"  # Modifies an existing item
```

2 Tuple

An ordered collection defined using parentheses `()`.

- **Immutable:** Cannot be modified, added to, or shortened once created.
- **Indexed:** Items are retrieved using numerical indices just like lists.
- **Best used for:** Fixed, read-only data that should remain constant throughout a program (e.g., GPS coordinates, RGB color codes).

```
location = (37.7749, -122.4194)  # (Latitude, Longitude)
latitude = location[0]            # Accesses 37.7749
```

3 Dictionary

An ordered collection defined using curly braces {} that stores data as `key: value` pairs.

- **Key-Based Access:** Values are accessed by unique labeled keys rather than numeric indices.
- **Mutable Values:** You can modify values or insert new key-value pairs at any time.
- **Best used for:** Structured data, profiles, and records where attributes are named (e.g., user profiles, product listings).

```
user = {
    "username": "alex99",
    "email": "alex@example.com",
    "age": 25
}
print(user["email"])  # Retrieves 'alex@example.com'
```

4 At-a-Glance Comparison

Feature	List	Tuple	Dictionary
Syntax	`[item1, item2]`	`(item1, item2)`	`{"key": "value"}`
Modifiable?	Yes	No	Yes (values can change)
Lookup	By index (`list[0]`)	By index (`tuple[0]`)	By custom key (`dict["key"]`)
Duplicates	Allowed	Allowed	Keys unique; values duplicate

Table 1: Comparison of core Python data structures.